

Pradeep Singh Ramola

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TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, C++, TypeScript, SQL
Frameworks & Libraries: .NET Core, Angular, React, Spring Boot, Ext JS, PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn, Pandas, NumPy
AI / ML: LLMs, Fine-Tuning (LoRA/PEFT), RAG, Agentic AI, NLP, Computer Vision, Transformer Architectures, LangGraph, LangChain, Vector Databases, Optuna
Databases: Microsoft SQL Server, PL/SQL, PostgreSQL, Oracle 12c, MongoDB
Cloud & DevOps: Azure, AWS, Docker, Git, SVN, CI/CD, REST APIs, Microservices, Postman, Jira
Tools & IDEs: Visual Studio, VS Code, Sencha Cmd, Jupyter Notebook, PyCharm, SSMS, PL/SQL Developer
Practices: Agile, Scrum, TDD, Code Review, RCA, KT Sessions, System Design

EDUCATION

University at Buffalo, SUNY

Aug 2025 – Dec 2026

M.S. Computer Science GPA: 3.77/4.0

Courses: Algorithms, Machine Learning, Deep Learning, Data Intensive Computing

KJ Somaiya Institute of Engineering and Information Technology

Jun 2017 – Jul 2021

B.E. Information Technology GPA: 9.05/10.0

EXPERIENCE

NEC Software Solutions

Jun 2021 – Jul 2025

Full Stack Software Engineer

- Resolved **30+** high-priority production incidents by reproducing issues, identifying code-level fixes, supporting RCA discussions, collaborating in daily stand-ups, and participating in cross-team KT sessions; reduced mean time to resolution (MTTR) by **20%** and post-release defects by **35%**.
- Optimized **5+** backend batch workflows processing **10,000+** daily operational records, cutting runtimes from **3 days to 5 hours**; developed **20+** API endpoints and UI screens improving system reliability and reducing customer-reported issues by **25%**.
- Developed **8+** core modules including Activity Requests, Activity Request Authorisation, Administration, Overtime Generation, TOIL Balance Calculation, and Payment Calculation; added new business rules to system function parameters reducing manual configuration effort by **30%**.
- Built and optimized web and mobile screens using Ext JS, .NET Framework, and PL/SQL; developed interface tools to export application data to CSV/Excel for **50,000+** officers across **20+** UK police forces, eliminating manual data extraction.
- Led end-to-end development of a Leave Management System serving **20,000+** employees using Angular, .NET Core, SQL Server, and Azure AD, replacing **100%** of manual workflows and reducing processing time from weeks to **2–3 days**.
- Integrated CI/CD pipelines cutting deployment effort by **40%**; led **15+** sprint demos and client calls, conducted **50+** code reviews, and mentored **3** junior engineers on backend architecture and testing practices.
- Led CARM upgrades 4.3 → 5.3/5.6 — architecting **10+** modules, refactoring **40K+** lines of scheduling logic, and resolving **100+** critical bugs, reducing module load times by **30%** and improving cross-team maintainability.

PROJECTS

TinyGPT: Character-Level GPT | Python | PyTorch | Transformers

- Built a **145K-parameter** decoder-only GPT from scratch with causal multi-head self-attention, pre-norm blocks, GELU MLPs, residual connections, positional embeddings, and tied input/output embeddings.
- Trained on TinyShakespeare with AdamW, cosine LR scheduling, gradient clipping, and checkpointing; reduced validation loss from **3.185 to 2.725** and implemented temperature-based autoregressive generation.

Vision Transformer (ViT): CIFAR-10 Classification | Python | PyTorch | Computer Vision

- Implemented ViT from scratch with convolutional patch embeddings, learnable CLS token, multi-head self-attention, pre-norm blocks, and classifier head for class-balanced CIFAR-10 experiments.
- Ran ablation studies across patch sizes, model depths, and positional encodings; built attention-entropy and spatial-correlation interpretability metrics to analyze patch-level attention behavior.

BERT Fine-Tuning: Hate Speech Classification | Python | PyTorch | Hugging Face | NLP

- Fine-tuned BERTweet on THOS Twitter dataset for 3-class classification; improved Test Macro F1 from **0.6486 to 0.6848** using domain adaptation, focal loss, and cosine LR scheduling.
- Applied LoRA to cut trainable parameters from **109M to 297K** (0.27%), retaining **98.3%** of full fine-tuning performance; ran fairness analysis and attention-based explainability across target groups.

PUBLICATIONS

Product Analyst Advisor: NLP and sentiment analysis for mobile product research. *IRJET*, Vol. 10, Issue 10, Oct 2023

HACKATHONS

- IBM Hackathon (Apr 2026):** Built Victor AI, an agentic AI assistant for international students using IBM watsonx Orchestrate, multi-agent workflows, email/calendar tools, and web scraping across 50+ universities.
- UB Hacking (Nov 2025):** Built BrailleSense, a real-time speech/text/video to Braille translator app for accessibility.